

Third Terminal Examination - 2079

Class : XI

Subject: Physics (1011)

Full Marks: 75

Time: 3 hrs.

SET 'B'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group A

Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]

- Whatever may be the directions of forces 4N and 2N, their resultant can't be
 - 1N
 - 2N
 - 3N
 - 4N
- Which of the following is always conserved in collision?
 - Kinetic energy
 - Angular momentum
 - Linear momentum
 - Torque
- An athlete makes a long jump and follows a projectile motion. Air resistance is negligible. Which one of the following statements is true about the athlete?
 - The athlete has a constant horizontal and vertical velocities.
 - The athlete has a constant horizontal velocity and constant downward acceleration
 - The athlete has a constant upward acceleration followed by a constant downward acceleration.
 - The athlete has a constant upward velocity followed by a constant downward velocity
- Which of the following is dimensionless number?
 - frequency
 - power of lens
 - strain
 - Planck's constant

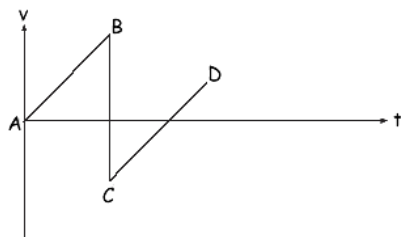
5. A force of 50N is just able to move a block of wood of mass 10 kg on a rough horizontal surface. It's coefficient of friction is
a. 1 b. 0.7 c. 0.5 d. 0.3
6. Lengths of two metal bars 'A' and 'B' are 8cm and 12cm respectively. If they show same difference in length at all temperature, their superficial expansivities are in the ratio
a. 2:3 b. 3:2 c. 9:4 d. 4:9
7. A concave mirror has radius of curvature 20 cm. An object is placed 10 cm from the pole of the mirror. The image will be at
a. 10 cm from the pole on same side
b. infinity
c. 20 cm from the pole on same
d. will not be formed
8. A beam of monochromatic blue light of wavelength 420 nm in air travels in water of refractive index 1.33. Its wavelength in nm in water will be
a. 560 nm b. 400 nm c. 315 nm d. 280 nm
9. Angle of deviation when passing through a prism is greatest for light:
a. red b. violet c. blue d. yellow
10. A charge 'q' is placed at the centre of the line joining the equal charges 'Q'. The system of three charges will be in equilibrium if q is equal to
a. $-Q/2$ b. $-Q/4$ c. $-2Q$ d. $Q/4$
11. Electric field intensity at a point due to an infinite sheet of charge having surface charge density ' σ ' is E . If sheet were conducting, electric intensity would be
a. $E/2$ b. E c. $2E$ d. $4E$

Group – B

Give short answer to the following questions.

[8 × 5 = 40]

12. a. State parallelogram law of vector addition. [1]
b. If a vector has zero magnitude, is it meaningful to call vector? [1]
c. A river 1 km wide is flowing at 3 km/hr. A swimmer whose velocity in still water is 4 km/hr wishes to cross the river straight.
i. In what direction should he strike out to reach the other bank. [1]
ii. What is the time taken by him to cross the river. [2]
13. a. When catching a ball, the cricketer moves his hands for a short distance in the direction of travel of the ball as it makes contact with his hands. Explain, why this technique results in saving his hands from possible injury. [2]
b. A cricket ball of mass 145 gm is moving with a velocity of 14m/s and is being hit by a bat, so that the ball is turned back with a velocity of 22m/s. The force of blow acts on the ball for 0.015 seconds. Find the average force exerted by bat on ball. [2]
c. Newton's first law of motion is contained in second law. Explain [1]
14. a. The diagram shows the velocity-time graph for a vertically bouncing ball, which is released above the the ground at A and strikes the floor at B. The effects of air resistance have been neglected.



- i. What does the gradient of a velocity-time graph represent? [1]
 - ii. Explain why the gradient of the line CD is the same as line AB. [1]
 - iii. What does the area between the line AB and the time axis represent? [1]
 - b. Show that there are two different angles of projection under same initial speed for the same horizontal range. [2]
15. a. State Newton's law of cooling. [1]
- b. A copper calorimeter weighing 53.2 g is first filled with water then with a liquid. Time taken to cool from 40°C to 32°C in the two cases is 4 mins and 3 mins respectively. The mass of water is 25 g and that of liquid is 30 g.
- i. Calculate the rate of heat lost by water and calorimeter.
 - ii. Calculate the rate of heat lost by liquid and calorimeter.
 - iii. Calculate the specific heat of the liquid. (The specific heat of copper = $0.094 \text{ cal/g } ^{\circ}\text{C}$.) [1+1+2]
16. a. A student claimed that thermometers are useless because a thermometer always registers its own temperature. How would you respond? [2]
- b. A copper vessel with a volume of exactly 100m^3 at a temperature of 15°C is filled with glycerin. If the temperature rises to 25°C , how much glycerin will spill out? [Given: α for copper $16.7 \times 10^{-6}/^{\circ}\text{C}$, γ for glycerin = $5.3 \times 10^{-4}/^{\circ}\text{C}$] [3]

OR

- a. You are looking to buy a small piece of land in Kathmandu. The price is “only” Rs 60,000 per square meter. The land title says the dimensions are $20\text{m} \times 30\text{m}$. By how much would the total price change if you measured the parcel with a steel tape measure on a day when the

temperature was 20°C above the temperature that the tape measure was designed for? The dimensions of the land do not change. (linear expansivity for steel is $12 \times 10^{-6}/\text{K}$). [3]

b. Why do we feel cold when we spray perfume on our body? [2]

17. a. On what factors does the lateral shift depend? [1]

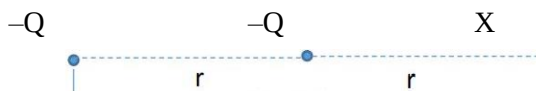
b. Calculate the critical angle for (i) an air glass surface (ii) an air water surface (iii) water glass surface. (${}_a\mu_g = 1.5$, ${}_a\mu_w = 1.33$) [3]

c. Does focal length change when the curve mirror is dipped in liquid? Why [1]

18. a. What are grazing incidence and grazing emergence? [2]

b. Derive an expression for the refractive index of the material of the prism in terms of the refracting angle and the angle of minimum deviation. [3]

19. a. Sketch an electric field pattern around two identical negative point charges shown below. [2]



b. Obtain an equation, in terms of Q and r , for the field strength at point X due to two charges shown in above figure. [3]

OR

a. If the distance between two equal point charges is doubled and their individual charges are also doubled what would happen to the force between them? [2]

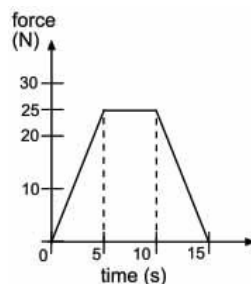
- b. Consider a uniform electric field, $E = 3 \times 10^3 \hat{i}$ N/C. [3]
- What is the flux of this field through a square of 10 cm on a side whose plane is parallel to the Y-Z plane?
 - What is the flux through the same square if the normal to plane makes a 60° angle with the X-axis?

Group – C

Give long answer to the following questions.

[3 × 8 = 24]

20. a. A force is applied to a stationary trolley to push across a horizontal floor. Force time graph is shown in the figure alongside.



- Calculate the magnitude of impulse that acts on trolley. [2]
 - The trolley reaches a velocity of 2.2 m/s after force is applied. Assume friction is negligible, calculate the mass of trolley. [2]
 - Explain why trolley continues to move at constant velocity after the force has been removed? [2]
- b. If the displacement of body is proportional to the square of time, state the nature of motion of body. [2]

OR

- Define angle of repose. How will you determine the coefficient of limiting friction between body and inclined plane? [1+2]
- Two tall buildings are 40m apart. With what speed must a ball be thrown horizontally from a window 120 m above ground in one building so that it will enter a window 40 m above ground in the other building. [3]

- c. Define significant figures. Find the number of significant figures in measurement 5.000 cm. [1+1]

21. a. In a report of an experiment to measure the specific latent heat of vaporisation of water, a student wrote the following.

“Steam at 100 °C was added to cold water in a calorimeter.

When the steam had condensed, measurements were taken.

The specific latent heat of vaporisation of water was then calculated.”

- i. Draw a labelled diagram of the apparatus used. [1]
 - ii. List two measurements that the student took before adding the steam to the water. [1]
 - iii. How did the student find the mass of steam that was added to the water? [1]
 - iv. How did the student make sure that only steam, and not hot water, was added to the calorimeter? [1]
 - v. Give one precaution that the student took to prevent heat loss from the calorimeter. [1]
- b. A tray containing 0.20 kg of water at 20 °C is placed in a freezer.

The temperature of the water drops to 0 °C in 10 minutes.

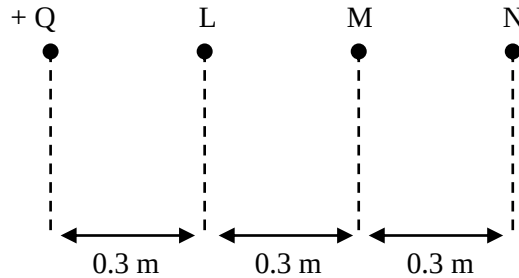
(specific heat capacity of water = $4200 \text{ J kg}^{-1} \text{ K}^{-1}$)

Calculate

- i. the energy lost by the water as it cools to 0°C.
 - ii. the average rate at which the water is losing energy, in J s^{-1} . [1+2]
22. a. Why do we sometimes experience a slight shock when getting out of a car? [1]
- b. Derive the expression for electric potential at a point due to a point charge. [3]

c. Observe the figure and answer the questions:

[3]



- i. If the electric potential at point L is $+3\text{V}$, find the magnitude of charge Q .
 - ii. Calculate the potential at point N.
- d. Is it possible to have a point where potential is zero but electric field non-zero?

[1]

☞ - The End - ☞